Lessons 7–10. Linear ODE Systems, Phase Portraits, Nonlinear Systems, and Lyapunov Functions

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1 Linear ODE Systems and the Matrix Exponential

1.1 From scalar to vector equations

Recall the scalar first-order linear ODE with constant coefficient:

$$\dot{x}(t) = ax(t), \qquad a \in \mathbb{R}.$$

Its general solution is $x(t) = Ce^{at}$. Now we want to extend this idea to a *system* of n equations in n unknown functions $x_1(t), \ldots, x_n(t)$. Writing the unknowns as a column vector $x(t) = (x_1(t), \ldots, x_n(t))^\top$, a first-order linear system with *constant matrix* $A \in \mathbb{R}^{n\times n}$ takes the compact form

$$\dot{x}(t) = Ax(t). \tag{6}$$

The analogy with the scalar case suggests that the solution should be $x(t) = e^{At}C$ for some constant vector $C \in \mathbb{R}^n$ — but we first need to make sense of the exponential of a *matrix*.

1.2 The matrix exponential

Definition 1.1 (Matrix exponential)**.** Let $A \in \mathbb{R}^{n\times n}$. The **matrix exponential** of A is defined by the convergent power series

$$e^{At} = \sum_{k=0}^{\infty} \frac{(At)^k}{k!} = I + At + \frac{A^2t^2}{2!} + \frac{A^3t^3}{3!} + \cdots .$$

This series converges absolutely for every matrix A and every $t \in \mathbb{R}$, since $\|A^k\| \le \|A\|^k$ and the scalar exponential series converges.

Key properties (analogues of the scalar case):

1. $e^{A\cdot 0} = I$ (set $t = 0$: every term with $k \ge 1$ vanishes).
2. $\dfrac{d}{dt}e^{At} = Ae^{At}$ (differentiate term by term: $\sum_{k=1}^{\infty} \frac{kt^{k-1}A^k}{k!} = A\sum_{k=1}^{\infty} \frac{A^{k-1}t^{k-1}}{(k-1)!} = Ae^{At}$).
3. If $AB = BA$ then $e^{(A+B)t} = e^{At}e^{Bt}$.
4. $(e^{At})^{-1} = e^{-At}$.

Why does this solve the system? If $x(t) = e^{At}C$, then

$$\dot{x}(t) = \frac{d}{dt}\left(e^{At}C\right) = Ae^{At}C = Ax(t).\checkmark$$

Moreover, $x(0) = e^{A\cdot 0}C = IC = C$, so C plays the role of the initial condition.

Theorem 1.1. *The general solution of $\dot{x}(t) = Ax(t)$ is $x(t) = e^{At}C$, where $C \in \mathbb{R}^n$ is an arbitrary constant vector.*

1.3 Computing e^{At} directly from the definition

For simple matrices the series can be summed explicitly.

Example 1.1 (Diagonal matrix)**.** Let $A = \begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix}$.

Solution. Note that $A^k = \begin{pmatrix} 2^k & 0 \\ 0 & 5^k \end{pmatrix}$, so

$$e^{At} = \sum_{k=0}^{\infty} \frac{1}{k!} \begin{pmatrix} (2t)^k & 0 \\ 0 & (5t)^k \end{pmatrix} = \begin{pmatrix} e^{2t} & 0 \\ 0 & e^{5t} \end{pmatrix}.$$

In general, for a diagonal matrix $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$: $e^{Dt} = \operatorname{diag}(e^{\lambda_1 t}, \ldots, e^{\lambda_n t})$.

Example 1.2 (Exchange (flip) matrix)**.** Let $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

Solution. Observe: $A^2 = I$, $A^3 = A$, $A^4 = I$, ... In general, $A^{2k} = I$ and $A^{2k+1} = A$. Splitting the series into even and odd powers:

$$\begin{aligned} e^{At} &= \sum_{k=0}^{\infty} \frac{t^{2k}}{(2k)!} I \; + \; \sum_{k=0}^{\infty} \frac{t^{2k+1}}{(2k+1)!} A \\ &= \cosh(t)\, I + \sinh(t)\, A = \begin{pmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{pmatrix}. \end{aligned}$$

Example 1.3 (Rotation matrix)**.** Let $A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

Solution. Here $A^2 = -I$, $A^3 = -A$, $A^4 = I$, etc. Again split even and odd:

$$\begin{aligned} e^{At} &= \sum_{k=0}^{\infty} \frac{(-1)^k t^{2k}}{(2k)!} I \; + \; \sum_{k=0}^{\infty} \frac{(-1)^k t^{2k+1}}{(2k+1)!} A \\ &= \cos(t)\, I + \sin(t)\, A = \begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix}. \end{aligned}$$

Notice that this is a rotation matrix: the system $\dot{x} = Ax$ produces circular trajectories.

Example 1.4 (Jordan block)**.** Let $A = \begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix}$, where $\alpha \in \mathbb{R}$.

Solution. Write $A = I + N$ where $N = \begin{pmatrix} 0 & \alpha \\ 0 & 0 \end{pmatrix}$. Since I and N commute, property (iii) gives $e^{At} = e^{It} e^{Nt}$. Now $N^2 = 0$ (nilpotent!), so the series for e^{Nt} terminates:

$$e^{Nt} = I + Nt + 0 + \cdots = \begin{pmatrix} 1 & \alpha t \\ 0 & 1 \end{pmatrix}.$$

Therefore

$$e^{At} = e^t \cdot \begin{pmatrix} 1 & \alpha t \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} e^t & \alpha t e^t \\ 0 & e^t \end{pmatrix}.$$

1.4 Computing e^{At} via Hermite interpolation

For larger or more complicated matrices, computing e^{At} by summing the power series is usually impractical. The following theorem provides a systematic shortcut.

Theorem 1.2 (Matrix exponential via Hermite interpolation)**.** *Let m be the degree of the minimal polynomial of A, let $\lambda_1, \ldots, \lambda_k$ be the distinct eigenvalues of A, and let $m_1, \ldots, m_k$ be their multiplicities in the minimal polynomial. There exists a polynomial p of degree at most $m-1$ such that*

$$e^{At} = p(A).$$

This polynomial (the Hermite interpolation polynomial) is uniquely determined by the conditions

$$p^{(i)}(\lambda_j) = t^i e^{\lambda_j t} \qquad (j = 1, \ldots, k;\ i = 0, 1, \ldots, m_j - 1).$$

Narration. The idea is as follows: if λ is a simple eigenvalue, then $e^{At}v = e^{\lambda t}v$ for any eigenvector v. For a repeated eigenvalue we also need the derivatives in λ. The theorem says that we only need to know the value (and finitely many derivatives) of $e^{\lambda t}$ at each eigenvalue; the rest is determined by linearity.

Example 1.5 (Using Hermite interpolation)**.** Compute e^{At} for $A = \begin{pmatrix} 2 & 1 \\ -1 & 4 \end{pmatrix}$ (eigenvalue $\lambda = 3$ with multiplicity 2 in the minimal polynomial).

Solution. We seek $p(s) = a_0 + a_1 s$ (degree ≤ 1) such that

$$p(3) = e^{3t} \quad \text{and} \quad p'(3) = te^{3t}.$$

From the first condition: $a_0 + 3a_1 = e^{3t}$. From the second condition: $a_1 = te^{3t}$. Hence $a_0 = e^{3t} - 3te^{3t}$ and

$$e^{At} = p(A) = a_0 I + a_1 A = (1-3t)e^{3t} I + te^{3t} A = e^{3t}\left[(1-3t)I + tA\right] = e^{3t}\begin{pmatrix} 1-t & t \\ -t & 1+t \end{pmatrix}.$$

1.5 Solved linear systems

In the following problems we find the general solution $x(t) = e^{At}C$ by first computing the eigenvalues and eigenvectors of A.

Example 1.6 (2×2 diagonal system)**.** $\begin{cases} \dot{x}_1 = x_1 \\ \dot{x}_2 = -x_2 \end{cases}$

Here $A = \operatorname{diag}(1, -1)$, eigenvalues $\lambda_1 = 1$, $\lambda_2 = -1$. The general solution is

$$x_1(t) = C_1 e^t, \qquad x_2(t) = C_2 e^{-t}.$$

Example 1.7 (Rotation system)**.** $\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = -x_1 \end{cases}$

This is $A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, already computed above.

$$x_1(t) = C_1 \cos t + C_2 \sin t, \qquad x_2(t) = -C_1 \sin t + C_2 \cos t.$$

Example 1.8 (Real eigenvalues, 2×2). $\begin{cases} \dot{x}_1 = x_1 - x_2 \\ \dot{x}_2 = -4x_1 + x_2 \end{cases}$

$A = \begin{pmatrix} 1 & -1 \\ -4 & 1 \end{pmatrix}$. Characteristic equation: $(1-\lambda)^2 - 4 = 0 \Rightarrow \lambda^2 - 2\lambda - 3 = 0 \Rightarrow (\lambda - 3)(\lambda + 1) = 0$. Eigenvalues: $\lambda_1 = 3$, $\lambda_2 = -1$.

Eigenvectors:

- $\lambda_1 = 3$: $(A - 3I)v = 0 \Rightarrow \begin{pmatrix} -2 & -1 \\ -4 & -2 \end{pmatrix} v = 0 \Rightarrow v_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$.
- $\lambda_2 = -1$: $(A + I)v = 0 \Rightarrow \begin{pmatrix} 2 & -1 \\ -4 & 2 \end{pmatrix} v = 0 \Rightarrow v_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$.

General solution:

$$x(t) = C_1 e^{3t} \begin{pmatrix} 1 \\ -2 \end{pmatrix} + C_2 e^{-t} \begin{pmatrix} 1 \\ 2 \end{pmatrix},$$

i.e., $x_1(t) = C_1 e^{3t} + C_2 e^{-t}$, $x_2(t) = -2C_1 e^{3t} + 2C_2 e^{-t}$.

Example 1.9 (Complex eigenvalues). $\begin{cases} \dot{x}_1 = x_1 + x_2 \\ \dot{x}_2 = 3x_2 - 2x_1 \end{cases}$

$A = \begin{pmatrix} 1 & 1 \\ -2 & 3 \end{pmatrix}$. Characteristic equation: $(1-\lambda)(3-\lambda) + 2 = 0 \Rightarrow \lambda^2 - 4\lambda + 5 = 0 \Rightarrow \lambda = 2 \pm i$.

For $\lambda = 2 + i$: $(A - (2+i)I)v = 0$:

$$\begin{pmatrix} -1-i & 1 \\ -2 & 1-i \end{pmatrix} v = 0 \quad \Rightarrow \quad v = \begin{pmatrix} 1 \\ 1+i \end{pmatrix}.$$

The real and imaginary parts of $e^{(2+i)t}v$ give two real solutions:

$$e^{2t}\left[\cos t \begin{pmatrix} 1 \\ 1 \end{pmatrix} - \sin t \begin{pmatrix} 0 \\ 1 \end{pmatrix}\right] \quad \text{and} \quad e^{2t}\left[\sin t \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \cos t \begin{pmatrix} 0 \\ 1 \end{pmatrix}\right].$$

General solution:

$$x_1(t) = e^{2t}(C_1 \cos t + C_2 \sin t), \quad x_2(t) = e^{2t}\big(C_1(\cos t - \sin t) + C_2(\sin t + \cos t)\big).$$

Example 1.10 (Double eigenvalue, defective matrix). $\begin{cases} \dot{x}_1 = 2x_1 + x_2 \\ \dot{x}_2 = -x_1 + 4x_2 \end{cases}$

$A = \begin{pmatrix} 2 & 1 \\ -1 & 4 \end{pmatrix}$. Characteristic equation: $(2-\lambda)(4-\lambda) + 1 = 0 \Rightarrow \lambda^2 - 6\lambda + 9 = 0 \Rightarrow (\lambda - 3)^2 = 0$. Double eigenvalue $\lambda = 3$.

$(A - 3I) = \begin{pmatrix} -1 & 1 \\ -1 & 1 \end{pmatrix}$: only one eigenvector $v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Find a generalised eigenvector: $(A - 3I)v_2 = v_1$, i.e., $\begin{pmatrix} -1 & 1 \\ -1 & 1 \end{pmatrix} v_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \Rightarrow v_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ (for instance).

General solution via Jordan form:

$$x(t) = C_1 e^{3t} v_1 + C_2 e^{3t}(t v_1 + v_2) = e^{3t} \begin{pmatrix} C_1 + C_2 t \\ C_1 + C_2(1+t) \end{pmatrix}.$$

Hence

$$x_1(t) = e^{3t}(C_1 + C_2 t), \qquad x_2(t) = e^{3t}(C_1 + C_2 + C_2 t).$$

Example 1.11 (Lower triangular system). $\begin{cases} \dot{x}_1 = x_1 \\ \dot{x}_2 = x_1 + x_2 \end{cases}$

The first equation gives $x_1(t) = C_1 e^t$ immediately. Substituting into the second: $\dot{x}_2 = C_1 e^t + x_2$, which is a first-order linear ODE. Using variation of parameters ($x_2 = u(t)e^t$):

$$\dot{u}e^t + ue^t = C_1 e^t + ue^t \Rightarrow \dot{u} = C_1 \Rightarrow u = C_1 t + C_2.$$

Hence $x_2(t) = (C_1 t + C_2)e^t$.

2 Phase Portraits of Linear Systems

2.1 The phase plane

For a two-dimensional system $\dot{x} = Ax$ ($A \in \mathbb{R}^{2\times 2}$), instead of plotting $x_1(t)$ and $x_2(t)$ separately as functions of t, we can represent each solution as a **curve (trajectory)** in the (x_1, x_2)-plane, called the **phase plane**. The collection of all trajectories is the **phase portrait**.

This requires one fewer dimension and gives a global geometric picture of the system's behaviour.

2.2 Reduction to Jordan normal form

Let $P \in \mathbb{R}^{2\times 2}$ be invertible and set $y = Px$. Then $\dot{y} = P\dot{x} = PAx = PAP^{-1}y$, so the system becomes $\dot{y} = (PAP^{-1})y$. Since an invertible linear change of variables is a global one-to-one geometric deformation of the plane (stretch/rotation/shear), the *qualitative type* of the phase portrait is preserved. It therefore suffices to analyse the following three Jordan normal forms:

$$\text{(I)} \begin{pmatrix} \lambda & 0 \\ 0 & \mu \end{pmatrix} \qquad \text{(II)} \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix} \qquad \text{(III)} \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix}$$

2.3 Case I: Two real distinct eigenvalues

The general solution is $x_1(t) = C_1 e^{\lambda t}$, $x_2(t) = C_2 e^{\mu t}$ (assume $\lambda \neq \mu$, both nonzero). Eliminating t:

$$x_2 = C_2 \left(\frac{x_1}{C_1}\right)^{\mu/\lambda} = C \cdot x_1^{\mu/\lambda}.$$

Sub-case 1: $\lambda, \mu > 0$ **(Unstable node).** All trajectories diverge from the origin as $t \to +\infty$. If $0 < \mu < \lambda$, the x_1-axis (corresponding to the larger eigenvalue λ) is tangent to all trajectories at the origin.

Sub-case 2: $\lambda, \mu < 0$ **(Stable node).** All trajectories converge to the origin. The axes align with the eigenvectors of A.

Sub-case 3: $\lambda < 0 < \mu$ **(Saddle).** The trajectory equation $x_2 = Cx_1^{\mu/\lambda}$ (with $\mu/\lambda < 0$) gives hyperbola-like curves. The stable manifold (trajectories approaching the origin) lies along the eigenvector for λ; the unstable manifold along the eigenvector for μ.

2.4 Case II: Double eigenvalue, defective matrix

The solution is $x_1(t) = (C_1 + C_2 t)e^{\lambda t}$, $x_2(t) = C_2 e^{\lambda t}$. For $C_2 \neq 0$ and $\lambda \neq 0$:

$$x_2 = C_2 e^{\lambda t} \quad \Rightarrow \quad t = \frac{1}{\lambda} \ln \left| \frac{x_2}{C_2} \right|.$$

Substituting into $x_1 = (C_1 + C_2 t)e^{\lambda t}$ gives

$$x_1 = x_2 \left(\frac{C_1}{C_2} + \frac{1}{\lambda} \ln \left| \frac{x_2}{C_2} \right| \right).$$

Sub-case 1: $\lambda > 0$ **(Unstable degenerate node).** Trajectories leave the origin as $t \to +\infty$, tangent to the unique eigenvector direction. There is no spiralling in this case.

Sub-case 2: $\lambda < 0$ **(Stable degenerate node).** Trajectories converge to the origin tangentially to the single eigenvector direction.

2.5 Case III: Complex eigenvalues $\alpha \pm \beta i$, $\beta \neq 0$

Switch to polar coordinates $x_1 = r\cos\varphi$, $x_2 = r\sin\varphi$. The system in Case III becomes

$$\dot{x}_1 = \alpha x_1 - \beta x_2, \qquad \dot{x}_2 = \beta x_1 + \alpha x_2.$$

Differentiate $r^2 = x_1^2 + x_2^2$:

$$2r\dot{r} = 2x_1\dot{x}_1 + 2x_2\dot{x}_2 = 2x_1(\alpha x_1 - \beta x_2) + 2x_2(\beta x_1 + \alpha x_2) = 2\alpha(x_1^2 + x_2^2) = 2\alpha r^2,$$

so $\dot{r} = \alpha r$. Differentiate $\tan\varphi = x_2/x_1$:

$$\frac{\dot{\varphi}}{\cos^2\varphi} = \frac{\dot{x}_2 x_1 - x_2 \dot{x}_1}{x_1^2} = \frac{(\beta x_1 + \alpha x_2)x_1 - x_2(\alpha x_1 - \beta x_2)}{x_1^2} = \frac{\beta(x_1^2 + x_2^2)}{x_1^2} = \frac{\beta r^2}{x_1^2} = \frac{\beta}{\cos^2\varphi}.$$

Therefore $\dot{\varphi} = \beta$. Solving:

$$r(t) = r_0 e^{\alpha t}, \qquad \varphi(t) = \beta t + \varphi_0.$$

Sub-case 1: $\alpha > 0$ **(Unstable focus).** $r \to \infty$ while spiralling.

Sub-case 2: $\alpha < 0$ **(Stable focus).** $r \to 0$ while spiralling inward.

Sub-case 3: $\alpha = 0$ **(Center).** $r \equiv r_0$ — closed circular orbits (which become ellipses after inverse linear change of coordinates in the original variables).

2.6 The trace-determinant diagram

The type of the phase portrait can be read off from the **trace** $T = \operatorname{tr} A$ and **determinant** $D = \det A$ of the matrix, without computing eigenvalues explicitly. The characteristic equation is $\lambda^2 - T\lambda + D = 0$, with roots $\lambda_{1,2} = \frac{T \pm \sqrt{T^2 - 4D}}{2}$.

- $D < 0$: the roots are real with opposite signs $\Rightarrow$ **saddle**.
- $D > 0$, $T^2 - 4D > 0$: two real roots of the same sign $\Rightarrow$ **node** (stable if $T < 0$, unstable if $T > 0$).
- $D > 0$, $T^2 - 4D = 0$: double real root $\Rightarrow$ **degenerate node** or star node.
- $D > 0$, $T^2 - 4D < 0$: complex conjugate roots $\Rightarrow$ **focus** (stable if $T < 0$, unstable if $T > 0$) or **center** (if $T = 0$).

The parabola $T^2 = 4D$ separates nodes from foci.

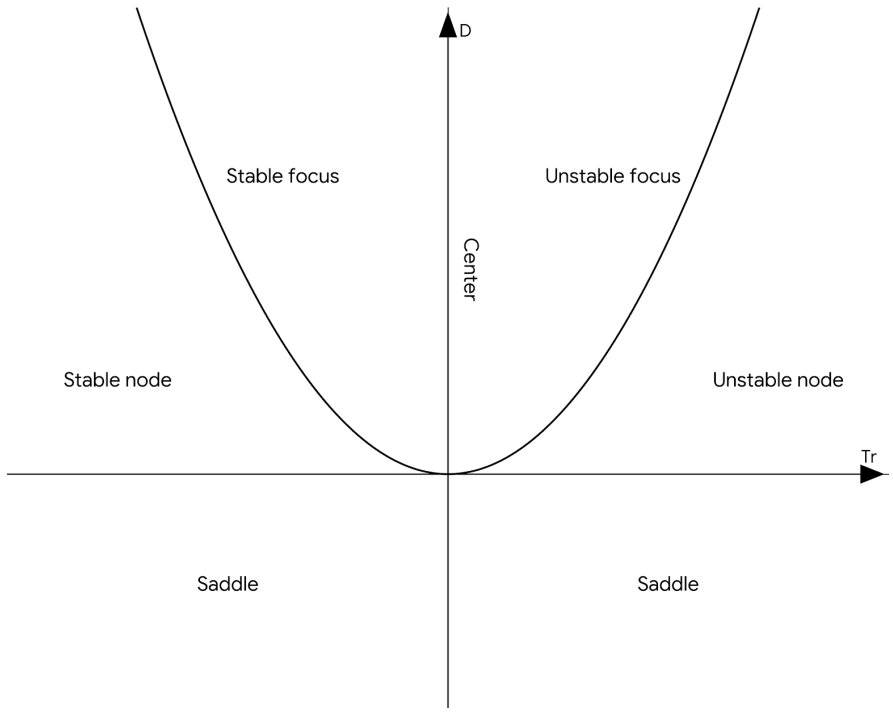

Figure 1: Classification of equilibrium types in the trace-determinant plane. The parabola $\mathrm{Tr}^2 = 4\det$ separates nodes (outside) from foci (inside).

2.7 Phase portrait exercises

Example 2.1 (Center)**.** $\begin{cases} \dot{x}_1 = -x_2 \\ \dot{x}_2 = x_1 \end{cases}$

$T = 0$, $D = 1 > 0$, $T^2 - 4D = -4 < 0$: centre. Eigenvalues $\pm i$. The solution is $x_1(t) = C_1 \cos t - C_2 \sin t$, $x_2(t) = C_1 \sin t + C_2 \cos t$. Every trajectory is a circle centred at the origin.

Example 2.2 (Focus)**.** $\begin{cases} \dot{x}_1 = \omega x_1 - x_2 \\ \dot{x}_2 = x_1 + \omega x_2 \end{cases}$

The matrix is $\begin{pmatrix} \omega & -1 \\ 1 & \omega \end{pmatrix}$. $T = 2\omega$, $D = \omega^2 + 1 > 0$. Eigenvalues $\omega \pm i$. In polar coordinates: $\dot{r} = \omega r$, $\dot{\varphi} = 1$.

- $\omega > 0$: unstable focus (outward spiral).
- $\omega = 0$: centre (circles).
- $\omega < 0$: stable focus (inward spiral).

Example 2.3 (Saddle)**.** $\begin{cases} \dot{x}_1 = x_1 - x_2 \\ \dot{x}_2 = -4x_1 + x_2 \end{cases}$

$T = 2$, $D = 1 - 4 = -3 < 0$: saddle. Eigenvalues $\lambda = 3, -1$ (computed earlier). The stable manifold is along $v_2 = (1, 2)^\top$; the unstable manifold along $v_1 = (1, -2)^\top$.

Example 2.4 (Unstable degenerate node)**.** $\begin{cases} \dot{x}_1 = 2x_1 + x_2 \\ \dot{x}_2 = -x_1 + 4x_2 \end{cases}$

$T = 6$, $D = 9$, $T^2 - 4D = 0$: degenerate (improper) node. Double eigenvalue $\lambda = 3 > 0$: unstable. Trajectories approach the origin tangent to the eigenvector $v = (1, 1)^\top$ (as $t \to -\infty$).

3 Autonomous and Nonlinear Systems

3.1 Autonomous systems and equilibrium points

An ODE system is called **autonomous** if the right-hand side does not depend explicitly on time:

$$\dot{x}(t) = f(x(t)), \qquad f : \mathbb{R}^n \to \mathbb{R}^n. \tag{7}$$

Constant solutions (also called **equilibrium** or **stationary** points): if $x(t) \equiv p$ for all t, then $\dot{x}(t) = 0$, so we need $f(p) = 0$. The equilibrium points are therefore the zeros of f.

Remark 3.1. For a general non-autonomous system $\dot{x} = f(t, x)$ this simplification is not available; for autonomous systems the phase portrait does not change with time, which makes global analysis much more tractable.

3.2 Linearisation near an equilibrium point

Let p be an equilibrium point of (7). Set $y(t) = x(t) - p$ (small perturbation). Then $\dot{y}(t) = \dot{x}(t) = f(x(t)) = f(p + y(t))$. By Taylor's theorem:

$$f(p + y) = f(p) + f'(p)\, y + r(y),$$

where $f'(p)$ is the Jacobian matrix of f at p and $r(y) = O(\|y\|^2)$ for $\|y\| \to 0$. Since $f(p) = 0$:

$$\dot{y}(t) = f'(p)\, y(t) + r(y(t)).$$

For small $\|y\|$ the remainder r is negligible compared to the linear term, so the local behaviour of (7) near p is governed by the **linearised system**

$$\dot{y}(t) = f'(p)\, y(t). \tag{8}$$

Theorem 3.1 (Hartman–Grobman / Linearisation theorem)**.** *Let $n = 2$, $f \in C^2$, and let p be an equilibrium point of (7). If every eigenvalue λ of the Jacobian $f'(p)$ satisfies* $\operatorname{Re}\lambda \neq 0$*, then the equilibrium point p of the nonlinear system (7) is of the* same type *(node, saddle, or focus) as the origin in the linearised system (8).*

Warning. If $\operatorname{Re}\lambda = 0$ (e.g. purely imaginary eigenvalues, giving a *centre* in the linear system), the theorem does not apply: the nonlinear system may have a centre, a stable focus, or an unstable focus — higher-order terms decide.

3.3 Definition of equilibrium types for nonlinear systems

We formalise what "same type" means. Write solutions near p in polar coordinates centred at p.

Definition 3.1. The equilibrium point p is called:

- a *stable focus* if $\lim_{t\to+\infty} r = 0$ and $\lim_{t\to+\infty} |\varphi| = \infty$;
- an *unstable focus* if $\lim_{t\to-\infty} r = 0$ and $\lim_{t\to-\infty} |\varphi| = \infty$;
- a *stable node* if $\lim_{t\to+\infty} r = 0$ and $\lim_{t\to+\infty} |\varphi| < \infty$;
- an *unstable node* if $\lim_{t\to-\infty} r = 0$ and $\lim_{t\to-\infty} |\varphi| < \infty$;
- a *saddle* if there exist exactly 2 trajectories approaching p as $t \to +\infty$ and 2 approaching as $t \to -\infty$, and all other nearby trajectories leave.

3.4 Nullclines and sketching phase portraits

To sketch the full phase portrait of an autonomous system $\dot{x} = f_1(x, y)$, $\dot{y} = f_2(x, y)$, proceed as follows:

1. Find all equilibrium points by solving $f_1 = 0$ and $f_2 = 0$ simultaneously.
2. Classify each equilibrium by computing $J = f'(p)$ and its eigenvalues.
3. Draw the **x-nullcline** $f_1 = 0$ and the **y-nullcline** $f_2 = 0$. These curves divide the phase plane into regions.
4. In each region, determine the sign of $\dot{x}$ and $\dot{y}$ by testing one point; this gives the direction (left/right, up/down) of the flow.
5. Combine the local behaviour near equilibria with the global direction information to sketch trajectories.

3.5 Worked examples

Example 3.1. $\begin{cases} \dot{x} = x - xy \\ \dot{y} = x^2 - y \end{cases}$

Step 1: Equilibria. $\dot{x} = 0$: $x(1-y) = 0$, so $x = 0$ or $y = 1$. $\dot{y} = 0$: $y = x^2$.

- $x = 0$, $y = 0$: equilibrium $(0, 0)$.
- $y = 1$, $y = x^2$: $x^2 = 1$, so $x = \pm 1$; equilibria $(1, 1)$ and $(-1, 1)$.

Equilibria: $(0, 0)$, $(1, 1)$, $(-1, 1)$.

Step 2: Jacobian.

$$J = \begin{pmatrix} 1-y & -x \\ 2x & -1 \end{pmatrix}.$$

At $(0, 0)$: $J = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, eigenvalues $1, -1$: **saddle**.

At $(1, 1)$: $J = \begin{pmatrix} 0 & -1 \\ 2 & -1 \end{pmatrix}$, $T = -1$, $D = 2$, $T^2 - 4D = 1 - 8 = -7 < 0$, $T < 0$: **stable focus**.

At $(-1, 1)$: $J = \begin{pmatrix} 0 & 1 \\ -2 & -1 \end{pmatrix}$, $T = -1$, $D = 2$, same discriminant: **stable focus**.

Example 3.2. $\begin{cases} \dot{x} = x(y-1) \\ \dot{y} = y(x-1) \end{cases}$

Equilibria. $\dot{x} = 0$: $x = 0$ or $y = 1$. $\dot{y} = 0$: $y = 0$ or $x = 1$. Possible pairs:

- $x = 0$, $y = 0$: $(0, 0)$.
- $x = 0$, $x = 1$: impossible.
- $y = 1$, $y = 0$: impossible.
- $y = 1$, $x = 1$: $(1, 1)$.

Equilibria: $(0, 0)$ and $(1, 1)$.

Jacobian.

$$J = \begin{pmatrix} y-1 & x \\ y & x-1 \end{pmatrix}.$$

At $(0, 0)$: $J = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$, eigenvalues $-1, -1$ (but $A = -I$, so $e^{At} = e^{-t}I$: all directions are equivalent). **Stable node** (actually a stable star node).

At $(1, 1)$: $J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, eigenvalues $+1, -1$: **saddle**.

Example 3.3. $\begin{cases} \dot{x} = y^2 - 4x^2 \\ \dot{y} = 8 - 4y \end{cases}$

Equilibria. $\dot{y} = 0$: $y = 2$. Then $\dot{x} = 0$: $4 - 4x^2 = 0$, so $x = \pm 1$. Equilibria: $(1, 2)$ and $(-1, 2)$.

Jacobian.

$$J = \begin{pmatrix} -8x & 2y \\ 0 & -4 \end{pmatrix}.$$

At $(1, 2)$: $J = \begin{pmatrix} -8 & 4 \\ 0 & -4 \end{pmatrix}$, eigenvalues -8 and -4 (triangular matrix): $T = -12 < 0$, $D = 32 > 0$, $T^2 - 4D = 144 - 128 = 16 > 0$: **stable node**.

At $(-1, 2)$: $J = \begin{pmatrix} 8 & 4 \\ 0 & -4 \end{pmatrix}$, eigenvalues 8 and -4 (opposite signs): **saddle**.

4 Lyapunov Functions and Lie Derivatives

4.1 Motivation

Consider the system

$$\dot{x} = -y - x^3, \qquad \dot{y} = x - y^3.$$

At the origin the Jacobian is $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, with eigenvalues $\pm i$. The linearised system is a centre — but the nonlinear terms may change this! To determine the true behaviour we use a **Lyapunov function**.

Let $V(x, y) = x^2 + y^2$. Along any solution $(x(t), y(t))$, define $V^*(t) = V(x(t), y(t))$. Differentiate:

$$\begin{aligned}\frac{dV^*}{dt} &= 2x\dot{x} + 2y\dot{y} = 2x(-y - x^3) + 2y(x - y^3) \\ &= -2xy - 2x^4 + 2xy - 2y^4 \\ &= -2(x^4 + y^4) < 0 \quad \text{for all } (x, y) \neq (0, 0).\end{aligned}$$

So $V^* = r^2$ is strictly decreasing along every non-zero trajectory: the origin is **asymptotically stable**. In particular, trajectories are driven toward the origin globally (and near the origin they rotate inward).

4.2 The Lie derivative

The calculation above can be formalised.

Definition 4.1 (Lie derivative)**.** Let $f : \mathbb{R}^n \to \mathbb{R}^n$ define the autonomous system $\dot{x} = f(x)$, and let $V : U \to \mathbb{R}$ be a differentiable function on an open set $U \subset \mathbb{R}^n$. The **Lie derivative** (or **derivative of V along the system**) is

$$L_f V := \langle \operatorname{grad} V,\, f \rangle = \partial_1 V \cdot f_1 + \partial_2 V \cdot f_2 + \cdots + \partial_n V \cdot f_n.$$

Key identity. If $x(t)$ is a solution, then

$$\frac{d}{dt} V(x(t)) = (L_f V)(x(t)).$$

Proof: By the chain rule, $\frac{d}{dt} V(x(t)) = \sum_i \partial_i V(x(t)) \cdot \dot{x}_i(t) = \langle \operatorname{grad} V(x(t)),\, f(x(t)) \rangle = (L_f V)(x(t))$. $\square$

Definition 4.2 (First integral)**.** If $L_f V \equiv 0$, then V is called a **first integral** of the system. This means $V(x(t))$ is constant along every solution.

4.3 Exercises on Lie derivatives

Example 4.1. Compute $L_f V$ for the system $\dot{x} = y$, $\dot{y} = -x - y$, and the functions:

(a) $V(x, y) = x^2 + y^2$.

$$L_f V = \frac{\partial V}{\partial x} \cdot y + \frac{\partial V}{\partial y} \cdot (-x - y) = 2x \cdot y + 2y \cdot (-x - y) = 2xy - 2xy - 2y^2 = -2y^2.$$

Since $L_f V = -2y^2 \leq 0$ (and equals zero only on the x-axis), the function V is non-increasing along solutions.

(b) $V(x, y) = \frac{3}{2}x^2 + xy + y^2$.

$$L_f V = (3x + y) \cdot y + (x + 2y)(-x - y) = 3xy + y^2 - x^2 - xy - 2xy - 2y^2 = -x^2 - y^2.$$

So $L_f V = -(x^2 + y^2) < 0$ for all $(x, y) \neq (0, 0)$: the origin is asymptotically stable.

Example 4.2 (Harmonic oscillator — first integral)**.** The harmonic oscillator $m\ddot{x} = -kx$ $(k > 0)$ written as a system:

$$\dot{x} = v, \qquad \dot{v} = -\frac{k}{m} x.$$

Total energy: $E(x,v) = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$.

Claim: E is a first integral, i.e., $L_f E \equiv 0$.

Proof:

$$L_f E = \frac{\partial E}{\partial x} \cdot v + \frac{\partial E}{\partial v} \cdot \left(-\frac{k}{m}x\right) = kx \cdot v + mv \cdot \left(-\frac{k}{m}x\right) = kvx - kvx = 0. \checkmark$$

The level curves $E = \text{const}$ are ellipses in the (x,v)-plane: every solution is periodic (no dissipation).

(a) Damped oscillator. Add drag: $\dot{v} = -\frac{k}{m}x + \frac{s(v)}{m}$ where $vs(v) < 0$ (drag opposes velocity). Then:

$$L_f E = kx \cdot v + mv \cdot \left(-\frac{k}{m}x + \frac{s(v)}{m}\right) = v\, s(v) < 0 \quad (\text{for } v \neq 0).$$

The energy strictly decreases along solutions: the oscillation gradually damps out, and the origin is asymptotically stable.

(b) Lotka–Volterra system. $\dot{x} = x - xy$, $\dot{y} = xy - y$ $(x, y \geq 0)$. Divide the two equations: $\frac{dy}{dx} = \frac{y(x-1)}{x(1-y)}$. Separate variables:

$$\frac{1-y}{y}\,dy = \frac{x-1}{x}\,dx \quad \Rightarrow \quad \int \left(\frac{1}{y} - 1\right) dy = \int \left(1 - \frac{1}{x}\right) dx \quad \Rightarrow \quad \ln y - y = x - \ln x + C.$$

Rearranging: $V(x,y) = x - \ln x + y - \ln y = C$ (first integral). The level curves of V are closed orbits surrounding the equilibrium $(1,1)$: the predator-prey populations oscillate periodically.

4.4 Lyapunov stability theorems

Let p be an equilibrium of $\dot{x} = f(x)$. A function $V : U \to \mathbb{R}$ defined on a neighbourhood U of p is called a **Lyapunov function** if:

1. $V(p) = 0$ and $V(x) > 0$ for all $x \in U \setminus \{p\}$ (*positive definite*),
2. V is continuously differentiable on U.

Theorem 4.1 (Lyapunov stability).

- *If $L_f V \leq 0$ on U, then p is **stable**.*
- *If $L_f V < 0$ on $U \setminus \{p\}$, then p is **asymptotically stable**.*
- *If $L_f V > 0$ on $U \setminus \{p\}$, then p is **unstable**.*

Interpretation. A positive definite V measures a "distance" from the equilibrium. If this distance is non-increasing along every solution ($L_f V \leq 0$), the system cannot escape a neighbourhood of p: the equilibrium is stable. If the distance strictly decreases, the system is drawn toward p.

4.5 Lyapunov function exercises

Example 4.3. Consider $\begin{cases} \dot{x} = -x + 5y^2 \\ \dot{y} = -3xy \end{cases}$ with $V(x, y) = 3x^2 + 5y^2$.

Check positive definiteness. $V(0, 0) = 0$ and $V(x, y) > 0$ for $(x, y) \neq (0, 0)$ since V is a sum of squares with positive coefficients. ✓

Compute $L_f V$.

$$\begin{aligned} L_f V &= \frac{\partial V}{\partial x}(-x + 5y^2) + \frac{\partial V}{\partial y}(-3xy) \\ &= 6x(-x + 5y^2) + 10y(-3xy) \\ &= -6x^2 + 30xy^2 - 30xy^2 = -6x^2 \leq 0. \end{aligned}$$

So $L_f V \leq 0$: the origin is **stable**. (It is not asymptotically stable because $L_f V = 0$ on the y-axis; the level curves $V = c$ are not all strictly contracted.)

Example 4.4. Consider $\begin{cases} \dot{x} = -x^3 - 2xy \\ \dot{y} = 2x^2 - 6y \end{cases}$ with $V(x, y) = x^2 + y^2$.

Compute $L_f V$.

$$L_f V = 2x(-x^3 - 2xy) + 2y(2x^2 - 6y) = -2x^4 - 4x^2y + 4x^2y - 12y^2 = -2x^4 - 12y^2 < 0$$

for $(x, y) \neq (0, 0)$. Hence the origin is **asymptotically stable**.

Example 4.5 (First integral and global phase portrait)**.** Sketch the global phase portrait of

$$\begin{cases} \dot{x} = -xy^4 \\ \dot{y} = x^6 y \end{cases}$$

using $V(x, y) = \dfrac{x^6}{6} + \dfrac{y^4}{4}$.

Compute $L_f V$.

$$L_f V = x^5 \cdot (-xy^4) + y^3 \cdot (x^6 y) = -x^6 y^4 + x^6 y^4 = 0.$$

So V is a **first integral**: each trajectory is contained in a level set

$$\frac{x^6}{6} + \frac{y^4}{4} = c.$$

Important narration: a closed level set does *not* automatically imply a periodic orbit. Here

$$x = 0 \quad \text{or} \quad y = 0 \quad \Rightarrow \quad \dot{x} = \dot{y} = 0,$$

so both coordinate axes consist entirely of equilibrium points.

For $xy \neq 0$, the signs are

$$\dot{x} = -xy^4 \text{ has sign } -\operatorname{sgn}(x), \qquad \dot{y} = x^6 y \text{ has sign } \operatorname{sgn}(y).$$

Hence in each quadrant, trajectories move monotonically along one arc of a level curve from an equilibrium on the x-axis to an equilibrium on the y-axis. Therefore the system has **no periodic orbits**; it has continua of equilibria plus connecting arcs.

Example 4.6 (Global asymptotic stability with polynomial Lyapunov function)**.** Consider

$$\dot{x} = xy^2 - x^3, \qquad \dot{y} = -y^3 - 2x^2y.$$

Take

$$V(x, y) = 2x^2 + y^2.$$

Then

$$L_f V = 4x(xy^2 - x^3) + 2y(-y^3 - 2x^2y) = 4x^2y^2 - 4x^4 - 2y^4 - 4x^2y^2 = -4x^4 - 2y^4 < 0$$

for every $(x, y) \neq (0, 0)$. Since V is positive definite and radially unbounded, the origin is **globally asymptotically stable**.

Example 4.7 (Semidefinite case and LaSalle argument)**.** Consider

$$\dot{x} = 2y^5 - x^3, \qquad \dot{y} = -2xy^2,$$

with

$$V(x, y) = 2x^2 + y^4.$$

We obtain

$$L_f V = 4x(2y^5 - x^3) + 4y^3(-2xy^2) = 8xy^5 - 4x^4 - 8xy^5 = -4x^4 \leq 0.$$

So Lyapunov's direct theorem gives stability. For asymptotic stability, inspect

$$\{(x, y) : L_f V = 0\} = \{x = 0\}.$$

On this set, $\dot{x} = 2y^5$, $\dot{y} = 0$, so the only invariant point is $(0, 0)$. By LaSalle's invariance principle, every trajectory tends to the origin. Thus the origin is **asymptotically stable** (indeed global, since V is radially unbounded).

5 Periodic Orbits: Existence and Non-existence

5.1 Motivation: why do we care about closed orbits?

So far we have classified equilibrium points and described how trajectories behave near them. But a two-dimensional autonomous system can also exhibit a qualitatively different type of long-run behaviour: a trajectory that *returns to its starting point* and repeats the same loop forever. This is a **periodic orbit** (or closed orbit), and it is the second fundamental building block of two-dimensional phase portraits after equilibria.

Why does this matter?

- **Oscillations.** Many physical and biological systems oscillate in steady state: a heartbeat, a population cycle, the current in an electronic oscillator. All of these correspond to a periodic orbit in the phase plane, not to an equilibrium.

- **Trapping.** In two dimensions, a closed orbit acts as an impassable wall: by the uniqueness theorem, no trajectory can cross a closed orbit, so the interior region is invariant. If we can show that the flow points inward on some closed curve, the trajectory must stay inside and — if there is no equilibrium inside — eventually approach a closed orbit.

- **Ruling out oscillations.** Equally important is knowing when oscillations *cannot* occur. The Bendixson criterion gives a simple divergence test: if div f has constant sign, there is no closed orbit anywhere in that region.

In this section we first define periodic solutions and limit cycles precisely, then state and apply the two main tools: the **Poincaré–Bendixson theorem** (existence) and the **Bendixson–Dulac criterion** (non-existence).

5.2 Periodic solutions and closed orbits

Definition 5.1 (Periodic solution)**.** A solution $x(t)$ of $\dot{x} = f(x)$ is called **periodic** if there exists a $T > 0$ such that $x(t+T) = x(t)$ for all $t \in \mathbb{R}$. The smallest such T is the **period**.

Definition 5.2 (Closed orbit)**.** A trajectory is a **closed orbit** if it forms a closed curve in the phase plane. A closed orbit is exactly the geometric trace of a periodic solution.

Remark 5.1. The existence and uniqueness theorem implies that no two distinct trajectories can intersect. Hence a trajectory cannot cross a closed orbit: interior and exterior are invariant regions. This topological property is one of the main reasons planar systems are tractable.

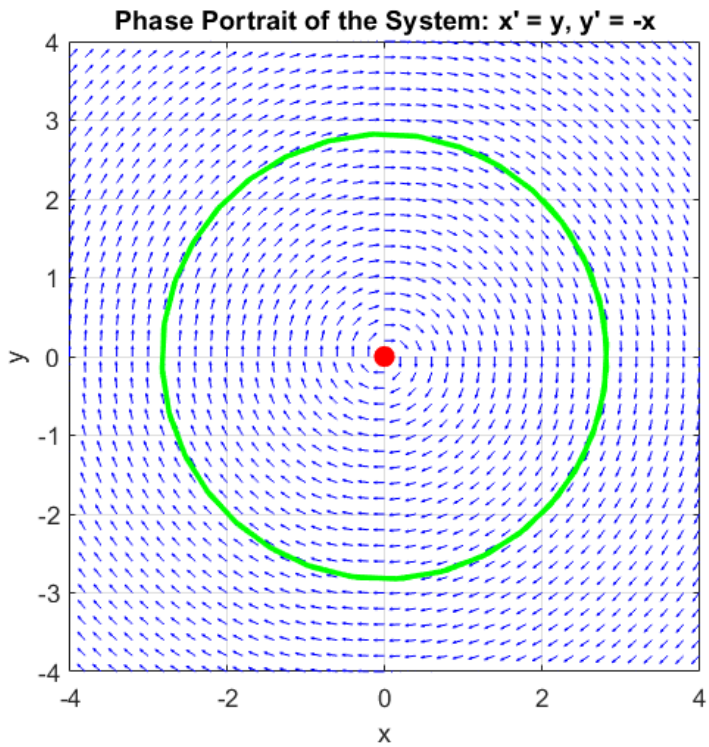

Figure 2: Left: a phase portrait without any closed orbit — all trajectories tend to an equilibrium. Right: a phase portrait with a closed orbit — the trajectory inside circles the equilibrium and the outer trajectory spirals toward the closed loop.

5.3 Limit cycles

Not all closed orbits are dynamically interesting. In a Hamiltonian system (like the harmonic oscillator), the whole phase plane is filled with closed orbits — a continuum. The interesting case is an **isolated** closed orbit, because then it attracts (or repels) nearby trajectories and represents a robust oscillation.

Definition 5.3 (Limit cycle)**.** A **limit cycle** is an *isolated* closed trajectory. Isolated means that nearby trajectories are not themselves closed; they spiral either toward or away from the limit cycle.

- **Stable** (attracting) limit cycle: all neighbouring trajectories spiral toward it as $t \to +\infty$.
- **Unstable** (repelling) limit cycle: all neighbouring trajectories spiral away as $t \to +\infty$ (approach it as $t \to -\infty$).
- **Half-stable**: trajectories on one side spiral toward it, on the other side away.

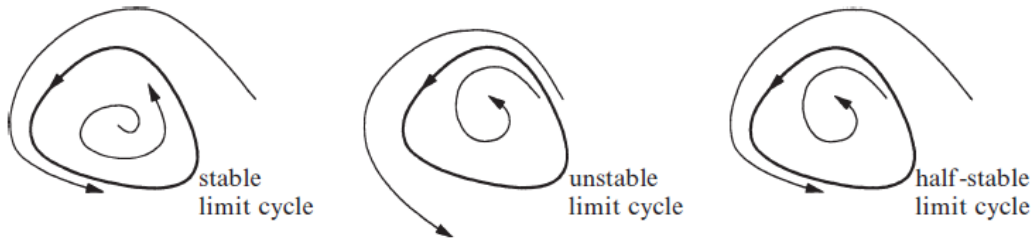

Figure 3: From left to right: stable limit cycle (surrounding trajectories are attracted), unstable limit cycle (surrounding trajectories are repelled), half-stable limit cycle.

Example 5.1 (A system with a stable limit cycle)**.** Consider (in polar coordinates) $\dot{r} = r(1-r^2)$, $\dot{\varphi} = 1$.

The circle $r = 1$ is an invariant set: if $r(0) = 1$ then $\dot{r} = 0$ always, so the trajectory stays on the circle and $\varphi(t) = t + \varphi_0$ winds around with period $T = 2\pi$.

For $r < 1$: $\dot{r} = r(1-r^2) > 0$, so r increases toward 1.

For $r > 1$: $\dot{r} = r(1-r^2) < 0$, so r decreases toward 1.

The circle $r = 1$ is therefore a **stable limit cycle**. Notice how the stability was determined just by checking the sign of $\dot{r}$ on each side — no eigenvalue calculation needed.

5.4 The Poincaré–Bendixson theorem

The main tool for proving the existence of a closed orbit. The idea: if we can find a closed, bounded region from which trajectories cannot escape, and the region contains no equilibrium, then the trajectory has nowhere to go except to wind around a closed orbit.

Theorem 5.1 (Poincaré–Bendixson)**.** *Suppose that:*

1. *R is a closed, bounded (compact) subset of the plane;*
2. *$\dot{x} = f(x)$ is a continuously differentiable vector field on an open set containing R;*

3. *R contains no equilibrium points;*

4. *there exists a trajectory C that is* confined *in R: it starts in R and remains in R for all future time $t \geq 0$.*

Then either C is itself a closed orbit, or C spirals toward a closed orbit as $t \to +\infty$. In either case, R contains at least one closed orbit.

Narration. The proof uses deep topology (the Jordan curve theorem), but the strategy is intuitive. The trajectory C is trapped in a bounded set. It cannot converge to an equilibrium (none exists in R). In two dimensions, the only other option is to accumulate on a closed curve.

How to apply Poincaré–Bendixson in practice.

1. Choose an annular region $R = \{(x, y) : r_1^2 \leq x^2 + y^2 \leq r_2^2\}$ (or any convenient region with a hole around the equilibrium).

2. Verify that the flow points *inward* on both the outer boundary and the inner boundary (or outward on both, giving trapping from outside).

3. Verify that the only equilibrium is inside the hole (so R itself has no equilibria).

4. Conclude: the trajectory starting in R stays in R and must approach a closed orbit.

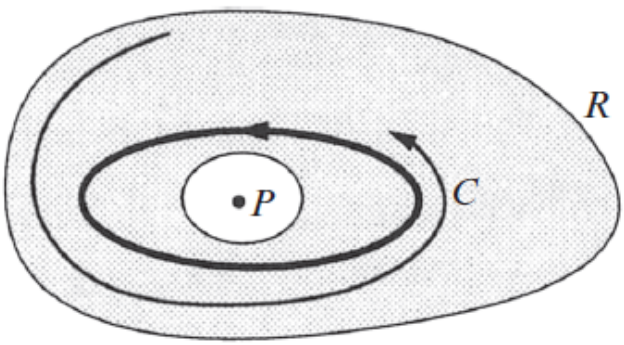

Figure 4: A trajectory confined in an annular region R with no equilibria. By Poincaré–Bendixson, there must be a closed orbit inside R.

Example 5.2 (Poincaré–Bendixson applied)**.** Consider the system

$$\begin{cases} \dot{x} = x - y - x(x^2 + y^2), \\ \dot{y} = x + y - y(x^2 + y^2). \end{cases}$$

Switch to polar coordinates ($r^2 = x^2 + y^2$).

Derive $\dot{r}$:

$$r\dot{r} = x\dot{x} + y\dot{y} = x\Big(x - y - xr^2\Big) + y\Big(x + y - yr^2\Big) = x^2 - xy - x^2r^2 + xy + y^2 - y^2r^2 = r^2 - r^4.$$

So $\dot{r} = r(1 - r^2)$.

Derive $\dot{\varphi}$: (One finds $\dot{\varphi} = 1$, the same calculation as the example above.)

The system in polar form is identical to the previous example! Thus:

- The circle $r = 1$ is an invariant closed orbit.
- For $r < 1$: $\dot{r} > 0$ (outward flow).
- For $r > 1$: $\dot{r} < 0$ (inward flow).

Take the annulus $R = \{\frac{1}{2} \le r \le 2\}$. On the inner circle $r = \frac{1}{2}$: $\dot{r} = \frac{1}{2}(1 - \frac{1}{4}) > 0$ (outward, so the inner boundary is crossed outward — trajectory enters R). On the outer circle $r = 2$: $\dot{r} = 2(1 - 4) < 0$ (inward — the outer boundary is also entered inward from the outside). The only equilibrium of the system is at the origin, which is *not* in R. By Poincaré–Bendixson, R contains a closed orbit. We already know it: $r = 1$.

5.5 Bendixson's criterion and Dulac's criterion

The main tools for proving the absence of closed orbits. The idea comes from Green's theorem (the divergence theorem in the plane): if C is a closed curve enclosing a region D, then

$$\oint_C (f\,dy - g\,dx) = \iint_D \left(\frac{\partial f}{\partial x} + \frac{\partial g}{\partial y}\right) dA.$$

Along a *solution* $x(t)$, $y(t)$ the left side equals $\oint_C (\dot{x}\,dy - \dot{y}\,dx)$ (since $\dot{x} = f$, $\dot{y} = g$). For a *closed orbit* traversed in time T:

$$\oint_C f\,dy - g\,dx = \int_0^T \Big(f(x(t), y(t))\,\dot{y}(t) - g(x(t), y(t))\,\dot{x}(t)\Big)\,dt = \int_0^T \Big(f\,g - g\,f\Big)\,dt = 0.$$

So the double integral of $\operatorname{div}(f, g) = \frac{\partial f}{\partial x} + \frac{\partial g}{\partial y}$ over the interior must be zero. If the divergence never changes sign, this is impossible — hence no closed orbit can exist.

Theorem 5.2 (Bendixson's criterion)**.** *Let $f, g \in C^1(R)$ for a simply connected domain $R \subset \mathbb{R}^2$. If the divergence*

$$\frac{\partial f}{\partial x} + \frac{\partial g}{\partial y}$$

is not identically zero and does not change sign in R, then the system $\dot{x} = f(x, y)$, $\dot{y} = g(x, y)$ has no closed orbit *lying entirely in R.*

Proof sketch. Suppose for contradiction that a closed orbit C with interior D lies entirely in R. By Green's theorem:

$$0 = \oint_C (f\,dy - g\,dx) = \iint_D \operatorname{div}(f, g)\,dA.$$

But if $\operatorname{div}(f, g)$ has constant sign and is not identically zero, the integral is nonzero — contradiction. □

Dulac's generalisation. The same argument applies if we multiply the vector field by a positive (or negative) function ψ before taking the divergence.

Theorem 5.3 (Dulac's criterion)**.** *Let $\psi \in C^1(R)$ be a function that does not change sign (and is not identically zero) in a simply connected domain R. If*

$$\frac{\partial(\psi f)}{\partial x} + \frac{\partial(\psi g)}{\partial y}$$

does not change sign and is not identically zero in R, then the system has no closed orbit *lying entirely in R.*

Narration. Bendixson is the special case $\psi \equiv 1$. Choosing a clever ψ can reveal the absence of closed orbits even when the plain divergence changes sign. A common choice is $\psi = x^a y^b$ or $\psi = e^{cx}$.

5.6 Worked examples

Example 5.3 (Bendixson's criterion: simple application)**.** Show that the system

$$\dot{x} = y, \qquad \dot{y} = -x - y + x^2$$

has no closed orbit in the entire plane.

Solution.

$$\operatorname{div}(f, g) = \frac{\partial y}{\partial x} + \frac{\partial(-x - y + x^2)}{\partial y} = 0 + (-1) = -1.$$

The divergence equals $-1 < 0$ everywhere (constant sign, nonzero). By Bendixson's criterion, there is no closed orbit.

Interpretation on the board. We computed a single derivative and got a negative constant. That is all we needed — no closed orbits, no limit cycles, anywhere in $\mathbb{R}^2$.

Example 5.4 (Bendixson's criterion with multiplier)**.** Show that the system

$$\dot{x} = y, \qquad \dot{y} = -ax - by + cx^2 + dy^2 \quad (b \neq 0)$$

has no closed orbit, using the Dulac multiplier $\psi(x, y) = e^{-2dx}$.

Solution. Compute:

$$\begin{aligned}
\frac{\partial(\psi f)}{\partial x} + \frac{\partial(\psi g)}{\partial y} &= \frac{\partial}{\partial x}\Big(e^{-2dx} y\Big) + \frac{\partial}{\partial y}\Big(e^{-2dx}(-ax - by + cx^2 + dy^2)\Big) \\
&= -2de^{-2dx} y + e^{-2dx}(-b + 2dy) \\
&= e^{-2dx}(-2dy - b + 2dy) = -b\, e^{-2dx}.
\end{aligned}$$

Since $e^{-2dx} > 0$ and $b \neq 0$, the expression $-be^{-2dx}$ has constant sign (negative if $b > 0$, positive if $b < 0$). By Dulac's criterion, there is no closed orbit.

Example 5.5 (Dulac to bound the number of limit cycles)**.** Consider

$$\begin{cases} \dot{x} = x - y - x(x^2 + y^2), \\ \dot{y} = x + y - y(x^2 + y^2). \end{cases}$$

(We already know $r = 1$ is a limit cycle.) Verify that no other limit cycle exists in the annulus $\frac{3}{4} < x^2 + y^2 < \frac{5}{4}$.

Solution. Compute the divergence with $\psi \equiv 1$:

$$\operatorname{div}(f, g) = \frac{\partial}{\partial x}\Big(x - y - x(x^2 + y^2)\Big) + \frac{\partial}{\partial y}\Big(x + y - y(x^2 + y^2)\Big).$$

$$= \Big(1 - (x^2 + y^2) - 2x^2\Big) + \Big(1 - (x^2 + y^2) - 2y^2\Big) = 2 - 2(x^2 + y^2) - 2(x^2 + y^2) = 2 - 4(x^2 + y^2).$$

In the annulus $\frac{3}{4} < r^2 < \frac{5}{4}$:

$$2 - 4 \cdot \tfrac{5}{4} = -3 < \operatorname{div}(f, g) < 2 - 4 \cdot \tfrac{3}{4} = -1 < 0.$$

The divergence is strictly negative throughout the annulus. Since the annulus is not simply connected, we do not apply Bendixson directly for non-existence. Instead, use a contradiction argument for uniqueness:

Assume there are two distinct closed orbits C_1 and C_2 in the annulus, and let A be the region between them. By Green's theorem,

$$\iint_A \operatorname{div}(f,g)\,dA = \oint_{\partial A}(f\,dy - g\,dx) = \oint_{C_2}(f\,dy - g\,dx) - \oint_{C_1}(f\,dy - g\,dx).$$

Each boundary integral is zero because along a trajectory $(\dot{x},\dot{y}) = (f,g)$, so $f\,\dot{y} - g\,\dot{x} = 0$. Hence $\iint_A \operatorname{div}(f,g)\,dA = 0$, contradiction with $\operatorname{div}(f,g) < 0$ on A. Therefore there is *at most one* closed orbit in the annulus.

Because we already know (from $\dot{r} = r(1-r^2)$) that $r = 1$ is a closed orbit, it follows that this closed orbit is unique in the annulus.

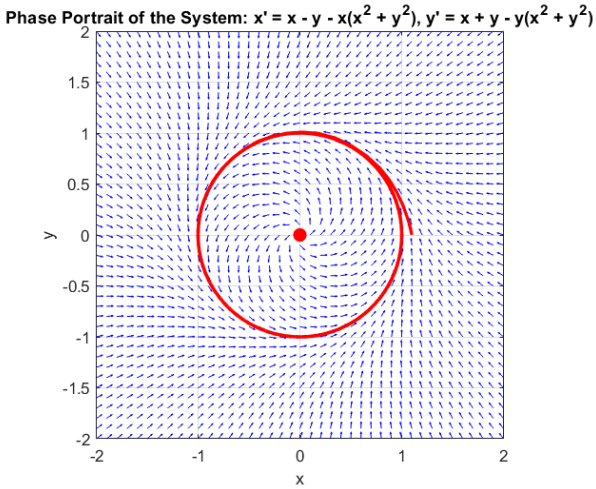

Figure 5: Phase portrait of the system: the unique limit cycle $r = 1$ attracting all trajectories from both inside and outside.

Example 5.6 (No closed orbits in a predator-prey variant)**.** Consider the system

$$\dot{x} = x(a - bx - cy), \quad \dot{y} = y(-d + ex), \qquad x, y > 0,$$

with $a, b, c, d, e > 0$. Use $\psi(x,y) = \dfrac{1}{xy}$ to show there are no closed orbits in the first quadrant $\{x > 0,\ y > 0\}$.

Solution. With $\psi = (xy)^{-1}$:

$$\frac{\partial(\psi f)}{\partial x} + \frac{\partial(\psi g)}{\partial y} = \frac{\partial}{\partial x}\left(\frac{a - bx - cy}{y}\right) + \frac{\partial}{\partial y}\left(\frac{-d + ex}{x}\right) = \frac{-b}{y} + 0 = -\frac{b}{y} < 0.$$

The expression is strictly negative for all $x, y > 0$. By Dulac's criterion, there are no closed orbits in the open first quadrant.

Remark. This system is a modified Lotka–Volterra model with logistic competition ($-bx$ term). The absence of closed orbits means the population either stabilises at an equilibrium or one species goes extinct — no perpetual oscillations.

5.7 Summary: the toolkit for two-dimensional autonomous systems

At this point we have all the main tools for analysing a planar autonomous system $\dot{x} = f(x,y)$, $\dot{y} = g(x,y)$:

1. **Find equilibria:** solve $f = g = 0$.
2. **Classify equilibria:** compute the Jacobian and its eigenvalues/trace/determinant; use the trace-determinant diagram.
3. **Look for closed orbits:**
   - If $\operatorname{div}(f, g)$ (or $\operatorname{div}(\psi f, \psi g)$ for some $\psi > 0$) has constant sign: **no closed orbits** (Bendixson–Dulac).
   - If you can trap a trajectory in a compact region without equilibria: **a closed orbit exists** (Poincaré–Bendixson).
4. **Use Lyapunov functions** to determine global stability or to find first integrals that give the trajectories explicitly.
5. **Nullclines** give the direction of the flow in each region and help connect the local pictures into a global phase portrait.